# CP2.3 — Methods in Calculus

## 1. Improper Integrals

An integral is **improper** if either:

- One or both limits are infinite: $\int_1^\infty f(x)\,dx$
- The integrand has a discontinuity (vertical asymptote) within the interval: $\int_0^1 \frac{1}{\sqrt{x}}\,dx$

We evaluate by replacing the problematic limit with a parameter and taking a limit.

### 1.1 Infinite Limits

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$$

If this limit exists and is finite, the integral **converges**. Otherwise it **diverges**.

**Example:** Determine whether $\int_1^\infty \frac{1}{x^2}\,dx$ converges.

$$\int_1^t \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_1^t = -\frac{1}{t} + 1 \xrightarrow{t\to\infty} 1$$

Converges to 1.

**Example:** Determine whether $\int_1^\infty \frac{1}{x}\,dx$ converges.

$$\int_1^t \frac{1}{x}\,dx = [\ln x]_1^t = \ln t \xrightarrow{t\to\infty} \infty$$

Diverges.

## 1.2 The p-Test (Critical Result)

$$\int_1^\infty \frac{1}{x^p}\,dx \text{ converges if and only if } p > 1$$

$$\int_0^1 \frac{1}{x^p}\,dx \text{ converges if and only if } p < 1$$

**Proof for $\int_1^\infty x^{-p}\,dx$:**

For $p \neq 1$: $\int_1^t x^{-p}\,dx = \left[\frac{x^{-p+1}}{1-p}\right]_1^t = \frac{t^{1-p}-1}{1-p}$

- If $p > 1$: $1-p < 0$, so $t^{1-p} \to 0$ as $t \to \infty$. Converges to $\frac{1}{p-1}$.
- If $p < 1$: $1-p > 0$, so $t^{1-p} \to \infty$. Diverges.
- If $p = 1$: gives $\ln t \to \infty$. Diverges.

## 1.3 Discontinuities within Interval

$$\int_a^b f(x)\,dx = \lim_{\varepsilon\to 0^+} \int_a^{b-\varepsilon} f(x)\,dx \quad \text{if } f \text{ has singularity at } b$$

**Example:** Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

$$\int_{\varepsilon}^{1} x^{-1/2}\,dx = [2\sqrt{x}]_{\varepsilon}^{1} = 2 - 2\sqrt{\varepsilon} \xrightarrow{\varepsilon \to 0^+} 2$$

Converges to 2.

**Exam tip:** Always write the integral with a limit parameter, evaluate the definite integral, then explicitly take the limit. Don't just "put in $\infty$" — show the limiting process.

## 2. Mean Value of a Function

The **mean value** of $f(x)$ on the interval $[a, b]$ is:

$$\bar{f} = \frac{1}{b-a}\int_{a}^{b} f(x)\,dx$$

Geometrically, this is the height of a rectangle of width $(b-a)$ having the same area as the area under $f(x)$ from $a$ to $b$.

**Example:** Find the mean value of $f(x) = x^2 + 1$ on $[1, 4]$.

$$\bar{f} = \frac{1}{4-1}\int_{1}^{4}(x^2+1)\,dx = \frac{1}{3}\left[\frac{x^3}{3} + x\right]_{1}^{4}$$

$$= \frac{1}{3}\left[\left(\frac{64}{3} + 4\right) - \left(\frac{1}{3} + 1\right)\right] = \frac{1}{3}\cdot\frac{63}{3+1}\cdot\frac{1}{1}$$

$$= \frac{1}{3}\left(\frac{64}{3} + 4 - \frac{1}{3} - 1\right) = \frac{1}{3}\cdot(21+3) = \frac{1}{3}\cdot 24 = 8$$

Wait — let me recalculate: $\frac{64}{3} + 4 - \frac{1}{3} - 1 = \frac{63}{3} + 3 = 21 + 3 = 24$. So $\bar{f} = \frac{24}{3} = 8$.

**Example:** Find the mean value of $f(x) = \sin x$ on $[0, \pi]$.

$$\bar{f} = \frac{1}{\pi}\int_0^{\pi} \sin x\, dx = \frac{1}{\pi}[-\cos x]_0^{\pi} = \frac{1}{\pi}(1+1) = \frac{2}{\pi}$$

# 3. Differentiation of Inverse Trigonometric Functions

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

## 3.1 Proof for arcsin

Let $y = \arcsin x$, so $\sin y = x$. Differentiating implicitly:

$$\cos y \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

(Taking positive root since $-\pi/2 \le y \le \pi/2$ so $\cos y \ge 0$.)

## 3.2 Chain Rule Extensions

| Function | Derivative |
|---|---|
| $\arcsin\left(\frac{x}{a}\right)$ | $\frac{1}{\sqrt{a^2-x^2}}$ |
| $\arctan\left(\frac{x}{a}\right)$ | $\frac{a}{a^2+x^2}$ |
| $\arcsin(f(x))$ | $\frac{f'(x)}{\sqrt{1-(f(x))^2}}$ |

**Example:** Differentiate $y = \arctan(3x^2)$.

$$\frac{dy}{dx} = \frac{6x}{1+(3x^2)^2} = \frac{6x}{1+9x^4}$$

# 4. Integration Using Inverse Trig Functions

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C, \quad |x| < a$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

**Example:** Evaluate $\displaystyle\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$.

$a = 2$. $\left[\arcsin\frac{x}{2}\right]_0^1 = \arcsin\frac{1}{2} - \arcsin 0 = \frac{\pi}{6} - 0 = \frac{\pi}{6}$

**Example:** Evaluate $\displaystyle\int \frac{1}{9+x^2}\,dx$.

$a = 3$. $\frac{1}{3}\arctan\left(\frac{x}{3}\right) + C$

## 4.1 Completing the Square

When the denominator is a quadratic $ax^2 + bx + c$, complete the square first to write it in the form $(x-h)^2 + k^2$ or $k^2 - (x-h)^2$.

**Example:** Find $\displaystyle\int \frac{1}{x^2+4x+13}\,dx$.

Complete the square: $x^2 + 4x + 13 = (x+2)^2 + 9$.

Substitute $u = x + 2$:

$$\int \frac{du}{u^2 + 9} = \frac{1}{3} \arctan \frac{u}{3} + C = \frac{1}{3} \arctan \frac{x+2}{3} + C$$

**Example:** Find $\displaystyle\int \frac{1}{\sqrt{8 - 2x - x^2}}\, dx$.

Complete the square: $8 - 2x - x^2 = 9 - (x+1)^2$.
Substitute $u = x + 1$:

$$\int \frac{du}{\sqrt{9 - u^2}} = \arcsin \frac{u}{3} + C = \arcsin \frac{x+1}{3} + C$$

**Exam tip:** Always check whether the numerator is a constant or has a variable part. If the numerator contains $x$, split as $\frac{Ax+B}{ax^2+bx+c} = \frac{A}{2a} \cdot \frac{2ax+b}{ax^2+bx+c} + \frac{\text{remainder}}{ax^2+bx+c}$. The first part integrates as a logarithm, the second as arctan.

# 5. Partial Fractions — Repeated and Irreducible Quadratic Factors

## 5.1 Repeated Linear Factors

$$\frac{f(x)}{(ax+b)^2(cx+d)} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{cx+d}$$

**Example:** Decompose $\dfrac{3x+1}{(x-1)^2(x+2)}$.

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

Multiply through: $3x + 1 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$

$x = 1$: $4 = 3B \Rightarrow B = \frac{4}{3}$
$x = -2$: $-5 = 9C \Rightarrow C = -\frac{5}{9}$
$x = 0$: $1 = -2A + 2B + C \Rightarrow 1 = -2A + \frac{8}{3} - \frac{5}{9} \to 2A = \frac{8}{3} - \frac{5}{9} - 1 = \frac{24-5-9}{9} = \frac{10}{9} \to A = \frac{5}{9}$
Check: $A + C = 0$ (coefficient of $x^2$ must be 0) ✓

## 5.2 Irreducible Quadratic Factors

$$\frac{f(x)}{(ax^2+bx+c)(dx+e)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{dx+e}$$

where $ax^2 + bx + c$ does not factorise over $\mathbb{R}$.

**Example:** Decompose $\dfrac{5x^2+4}{(x^2+4)(x-1)}$.

$$\frac{5x^2+4}{(x^2+4)(x-1)} = \frac{Ax+B}{x^2+4} + \frac{C}{x-1}$$

Multiply through: $5x^2 + 4 = (Ax+B)(x-1) + C(x^2+4)$
$x = 1$: $9 = 5C \Rightarrow C = \frac{9}{5}$
Expand and compare coefficients:
$x^2$: $5 = A + C \Rightarrow A = 5 - \frac{9}{5} = \frac{16}{5}$
$x^0$: $4 = -B + 4C = -B + \frac{36}{5} \Rightarrow B = \frac{36}{5} - 4 = \frac{16}{5}$
Check $x^1$: $0 = B - A = 0$ ✓

# 6. Integration by Parts — Repeated Application (Tabular Method)

$$\int u\,dv = uv - \int v\,du$$

For integrals requiring repeated integration by parts (e.g., $\int x^n e^x \, dx$, $\int e^{ax} \sin(bx) \, dx$), the **tabular method** (also called the DI method) is efficient.

## 6.1 Tabular Method for Polynomial × Exponential

**Example:** Find $\displaystyle\int x^3 e^x \, dx$ using the tabular method.

| Derivatives (D) | Integrals (I) | Sign |
|---|---|---|
| $x^3$ | $e^x$ | $+$ |
| $3x^2$ | $e^x$ | $-$ |
| $6x$ | $e^x$ | $+$ |
| $6$ | $e^x$ | $-$ |
| $0$ | $e^x$ | $+$ |

Multiply diagonally with alternating signs:

$$\int x^3 e^x \, dx = x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C = e^x(x^3 - 3x^2 + 6x - 6) + C$$

## 6.2 Cyclic Integration by Parts

**Example:** Find $\displaystyle\int e^{2x} \cos 3x \, dx$.

Let $I = \int e^{2x} \cos 3x \, dx$.
First IBP: $u = e^{2x}$, $dv = \cos 3x \, dx$, so $v = \frac{\sin 3x}{3}$:

$$I = \frac{e^{2x} \sin 3x}{3} - \int \frac{2e^{2x} \sin 3x}{3} \, dx$$

Second IBP on remaining integral: $u = e^{2x}$, $dv = \sin 3x\,dx$, $v = -\frac{\cos 3x}{3}$:

$$\int e^{2x}\sin 3x\,dx = -\frac{e^{2x}\cos 3x}{3} + \int \frac{2e^{2x}\cos 3x}{3}\,dx = -\frac{e^{2x}\cos 3x}{3} + \frac{2I}{3}$$

Substitute back:

$$I = \frac{e^{2x}\sin 3x}{3} - \frac{2}{3}\left(-\frac{e^{2x}\cos 3x}{3} + \frac{2I}{3}\right)$$

$$I = \frac{e^{2x}\sin 3x}{3} + \frac{2e^{2x}\cos 3x}{9} - \frac{4I}{9}$$

$$\frac{13I}{9} = \frac{3e^{2x}\sin 3x + 2e^{2x}\cos 3x}{9}$$

$$I = \frac{e^{2x}(3\sin 3x + 2\cos 3x)}{13} + C$$

**Exam tip:** For cyclic IBP ($e^{ax}\sin bx$ type), always call the original integral $I$, apply IBP twice keeping the same choice of "u", then solve for $I$. Don't forget the constant of integration.

## 7. Summary of Integration Formulas

| Integrand | Result |
|---|---|
| $\frac{1}{\sqrt{a^2 - x^2}}$ | $\arcsin\left(\frac{x}{a}\right) + C$ |
| $\frac{1}{a^2 + x^2}$ | $\frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$ |
| $\frac{1}{\sqrt{x^2 + a^2}}$ | $\operatorname{arsinh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 + a^2}\right) + C$ |
| $\frac{1}{\sqrt{x^2 - a^2}}$ | $\operatorname{arcosh}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2 - a^2}\right) + C$ |
| $\frac{1}{a^2 - x^2}$ | $\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert + C$ |

# 8. Exam-Style Practice Questions

1. **[Improper integral]** Determine whether $\int_0^\infty xe^{-x}\,dx$ converges, and if so, find its value. (Hint: use integration by parts.)
2. **[Improper integral]** Show that $\int_0^1 \frac{1}{x^{2/3}}\,dx$ converges and find its value.
3. **[Mean value]** Find the mean value of $f(x) = e^{2x}$ over the interval $[0, 3]$.
4. **[Inverse trig integration]** Find $\int \frac{3}{\sqrt{16-9x^2}}\,dx$.
5. **[Completing the square]** Evaluate $\int_0^1 \frac{2x+3}{x^2+2x+5}\,dx$.
6. **[Partial fractions]** Show that $\int \frac{4x^2+1}{(x^2+1)(2x-1)}\,dx = \ln|2x-1| + \arctan x + C$.
7. **[IBP tabular]** Find $\int x^2 \sin x\,dx$.
8. **[Cyclic IBP]** Find $\int e^{-x}\sin 2x\,dx$.

**General exam tips for CP2.3:**

- Improper integrals: always write $\lim_{t\to\infty}$ explicitly and evaluate before taking the limit
- For inverse trig integrals, identify $a$ correctly: in $\frac{1}{a^2+x^2}$, $a$ is the square root of the constant
- Completing the square: write out every algebraic step — errors here lose all subsequent marks
- IBP cyclic: rearranging to get $I$ on one side is the key step — don't omit the $+C$ at the end